

Week 3: Simultaneous Competition

Best Responses, Nash Equilibrium, and Iterated Elimination

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Choose without observing the rival, but reason about the rival's incentives.

From Week 2 to Week 3

Week 2: Sequential Moves

A later player observes an earlier action before responding.

- Game trees
- Backward induction
- Credible responses
- Timing advantage

Week 3: Simultaneous Moves

Players choose without observing one another's current actions.

- Payoff matrices
- Best responses
- Nash equilibrium
- Dominated strategies

The key distinction is observation, not clock time.

Opening Decision: Capacity Competition

The Situation

Two competing firms simultaneously choose **High Capacity** or **Low Capacity**. Neither observes the other's decision before committing.

	Rival: High	Rival: Low
You: High	(2, 2)	(5, 1)
You: Low	(1, 5)	(4, 4)

What would you choose? What do you expect the rival to choose?

First payoff: row player (you)

Second payoff: column player (rival)

Today's Learning Outcomes

By the end of class, you should be able to:

- 1 identify each player's **best responses**;
- 2 find and interpret a **pure-strategy Nash equilibrium**;
- 3 recognise games with one, several, or no pure equilibria;
- 4 simplify a game through **iterated elimination** of strictly dominated strategies;
- 5 construct and defend a payoff matrix for a business situation.

Best Response: Conditional on the Rival's Choice

Definition

A player's **best response** is a strategy that gives the highest payoff, given the strategy chosen by the other player.

Row player:

- Hold each column fixed.
- Compare the first numbers vertically.
- Mark every highest payoff.

Column player:

- Hold each row fixed.
- Compare the second numbers horizontally.
- Mark every highest payoff.

Blue = row player's payoff

Red = column player's payoff

If the rival chooses this, what should I choose?

Nash Equilibrium: Mutual Best Responses

Definition

A **Nash equilibrium** is a strategy profile in which every player's strategy is a best response to the strategies chosen by the others.

- No player can gain by changing strategy **alone**.
- Hold the other player's strategy fixed when checking a deviation.
- A game may have one, several, or no **pure-strategy** Nash equilibria.
- Equilibrium means strategically stable, not necessarily efficient or desirable.

Look for cells where the best-response marks overlap.

Solving the Capacity Game

	Rival: High	Rival: Low
You: High	(2, 2)	(5, 1)
You: Low	(1, 5)	(4, 4)

- High is the best response to both rival choices for each firm.
- High is therefore a strictly dominant strategy for each firm.
- The best-response marks overlap at High–High.
- High–High is the unique Nash equilibrium.

Important

Low–Low gives both firms 4, but either firm can gain by switching alone to High and receiving 5.

Dominant Strategy Versus Nash Equilibrium

Dominant Strategy

A strategy is best against **every** strategy the rival might choose.

This is a property of one player's strategy.

Nash Equilibrium

Each player's selected strategy is best against the strategy actually selected by the rival.

This is a property of the full strategy profile.

Relationship

Every dominant-strategy equilibrium is a Nash equilibrium, but many Nash equilibria do not use dominant strategies.

One Pure Equilibrium: Price Competition

	Rival: Cut	Rival: Maintain
You: Cut	(2, 2)	(4, 1)
You: Maintain	(1, 4)	(3, 3)

- Cut is strictly dominant for both firms.
- Cut–Cut is the unique Nash equilibrium.
- Maintain–Maintain is better for both, but either firm can gain by cutting alone.

Lesson

Strategic stability and collective desirability are different ideas.

Two Pure Equilibria: Product Standards

	Partner: Standard A	Partner: Standard B
You: Standard A	(4, 3)	(1, 1)
You: Standard B	(1, 1)	(3, 4)

- Both coordinated outcomes are Nash equilibria.
- Neither player has a dominant strategy.
- Both prefer compatibility, but they disagree about which standard to adopt.

Prediction Problem

Equilibrium analysis identifies two stable outcomes but does not by itself determine which one will occur.

No Pure Equilibrium: Competitive Positioning

	Rival: Segment A	Rival: Segment B
You: Segment A	(1, 3)	(3, 1)
You: Segment B	(3, 1)	(1, 3)

- You prefer to target a different segment from the rival.
- The rival prefers to target the same segment as you.
- Every cell gives at least one player an incentive to change.
- There is no pure-strategy Nash equilibrium.

Week 4 introduces mixed strategies for situations in which predictable pure actions cannot be stable.

Iterated Elimination of Strictly Dominated Strategies

Core Idea

If one strategy always gives a lower payoff than another strategy, remove it and analyse the smaller game again.

- 1 Compare one player's complete strategies across every rival strategy.
- 2 Remove a **strictly dominated** strategy.
- 3 Reconsider the reduced game for both players.
- 4 Repeat until no further strict dominance is available.

Important

Eliminate an entire strategy, not an unattractive cell. Weak dominance requires more care.

Worked Example: Eliminate and Reconsider

	Left	Centre	Right
Up	(4, 3)	(3, 2)	(2, 1)
Middle	(3, 1)	(2, 4)	(1, 3)
Down	(1, 2)	(1, 1)	(0, 0)

- 1 Down is strictly dominated by Up.
- 2 After removing Down, Right is strictly dominated by Centre.
- 3 In the reduced game, Middle is strictly dominated by Up.
- 4 With Up remaining, the column player chooses Left.

Surviving Prediction

Up-Left, with payoffs (4, 3).

Application 1: A Real Pricing Battle

Group Task

Select two real competing firms that choose prices without observing one another's current decision.

Construct the game:

- Define two or three price strategies.
- Explain the payoff drivers.
- Allow asymmetric firms if appropriate.
- Build the payoff matrix.

Analyse the game:

- Mark best responses.
- Identify dominant strategies, if any.
- Identify all pure Nash equilibria.
- Compare the prediction with reality.

Which business assumption drives your prediction?

Application 2: Capacity or Product Features

Group Task

Select two real firms and model one important competitive choice.

Choose one dimension:

- Large or small capacity
- Compatible or differentiated features
- Broad or focused product range

Analyse the game:

- Construct and explain the matrix.
- Mark best responses and simplify if useful.
- State the prediction and managerial advice.

Keep one strategic dimension unless additional choices materially change the prediction.

Checklist for the Two Application Matrices

Build the Model

- Real players and feasible strategies
- Timing and observation stated
- Payoffs supported by business logic
- Asymmetry included when relevant

Solve and Interpret

- All best responses marked
- Every pure equilibrium identified
- Dominance or elimination justified
- Prediction compared with reality

Final Check

Can you explain the matrix, solve it, and translate the prediction into managerial advice?

Application Debrief

- 1 Did the firms have dominant strategies?
- 2 How many pure-strategy Nash equilibria did the game have?
- 3 Which payoff assumption created that equilibrium pattern?
- 4 Did asymmetry between the firms matter?
- 5 Did the prediction fit the observed market outcome?

Managerial Lesson

Best-response marking is mechanical; constructing and defending the model is the demanding part.

Optional Extension: Subscription Pricing Battle

Use If Time Allows, or Continue Next Class

Two streaming services simultaneously choose **Premium Price** or **Discount Price**. A discount may attract subscribers but lowers margin; premium pricing protects margin but may lose price-sensitive customers.

- 1 Construct and explain a plausible payoff matrix.
- 2 Mark both players' best responses.
- 3 Identify every pure-strategy Nash equilibrium.
- 4 If no pure equilibrium exists, explain the cycle of deviations.
- 5 State one assumption that could reverse the result.

Week 3: Main Takeaways

- 1 Simultaneous means choosing without observing the rival's relevant action.
- 2 A best response is conditional on the rival's strategy.
- 3 A Nash equilibrium is a profile of mutual best responses.
- 4 A game may have one, several, or no pure-strategy equilibria.
- 5 Strictly dominated strategies can be removed iteratively.
- 6 Equilibrium means stable, not necessarily efficient or desirable.
- 7 Managerial conclusions depend on defensible payoff assumptions.

Construct the matrix. Mark best responses. Find every stable outcome. Defend the business logic.

Exit Ticket and Looking Ahead

Exit Ticket

Explain the difference between a dominant strategy and a Nash equilibrium. Why can a Nash equilibrium be undesirable?

Before Week 4

Complete the assigned pre-reading on mixed strategies. Come ready to ask:

What should a manager do when being predictable makes the firm exploitable?

Supplementary: Formal Definition of Nash Equilibrium

Definition

Let $s^* = (s_1^*, \dots, s_n^*)$ be a strategy profile. It is a Nash equilibrium if, for every player i ,

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \text{for every available } s_i.$$

- Hold the strategies of all other players fixed.
- Player i cannot gain by changing strategy alone.
- Equality is allowed because a player may have several best responses.

Supplementary: More Than Two Players

The Logic Does Not Change

A Nash equilibrium requires every player's strategy to be a best response to the strategies chosen by all other players.

- Two-player matrices are used because they are easier to display and discuss.
- With three or more players, a two-dimensional matrix becomes cumbersome.
- Add a player only when that player's strategic choice materially changes the incentives or prediction.
- Stakeholders that mainly affect profits, costs, or demand can often be represented in the payoffs instead.

Use the smallest model that preserves the strategic interaction.

Supplementary: Pure Versus Mixed Strategies

Pure Strategy

Choose one action with certainty.

Example: always choose Premium Price.

Mixed Strategy

Randomise among actions according to specified probabilities.

Example: choose Premium with probability p .

Week 4 Preview

Randomisation can be rational when a predictable pure strategy would be exploited.

Supplementary: Common Errors in Matrix Analysis

Common Error	Correction
Compare the two numbers inside one cell.	Follow one player's payoff position and color across strategies while holding the rival's strategy fixed.
Find the largest number and call it dominant.	Dominance requires comparison against every rival strategy.
Stop after finding one equilibrium.	Check every cell for overlapping best responses.
Assume equilibrium is efficient.	Equilibrium means no profitable unilateral deviation, not maximum joint payoff.
Eliminate an unattractive cell.	Eliminate a complete dominated strategy, an entire row or column.